ESLA-**IMP SIMP-M1,2,3**

By the TIE review team

1.The PDF for the random variable Z is $f_Z(z)=\begin{cases} \frac{1}{6\sqrt{Z}}; & 0 < Z < 9 \\ 0; & otherwise \end{cases}$

What are (i) the mean (ii) the mean of the square, and (iii) the variance ofthe random variable Z?

2. Given the data in the following table,

k	1	2	3	4	5
x_k	2.1	3.2	4.8	5.4	6.9
$P(x_k)$	0.21	0.18	0.20	0.22	0.19

(i) Plot the PDF and the CDF of the discrete random variable X.

(ii) Write expressions for PDF and CDF using unit-delta functions and unit – step functions.

3. Define a random variable and briefly discuss the following terms associated with random variables. (i) Sample space (ii) Distribution Function (iii) Probability Mass Function (iv) Probability Density Function

4. Consider the function $f_K(x) = \frac{1}{6}; 0 \le x \le 6$ which is being used as a probability density function for a continuous random variable X.

a) Show that $f_K(x)$ is a valid PDF (b) Find the probability that X lies in the range of 3 to 5.b o

5. The CDF for the random variable X is

$$F_K(x) = \begin{cases} 1 - e^{-\frac{x^2}{4}} & ; x \ge 0 \\ 0; & x < 0 \end{cases}$$

Evaluate $P(1.1 < X \le 2.1)$

6. It is given that E[X] = 36.5 and that $E[X^2] = 1432.3$

(i) Find the standard deviation of X. (ii)If Y = 4X − 500, find the mean and variance of Y.

7. State and prove rank nullity theorem

8. Differentiate between Discrete random variable and Continuous randomvariable

9. Apply Gram scmidt orthogonalization process(GSOP) to the vectors $\beta_1=(1,1,1)$; $\beta_2=(1,-1,2)$; $\beta_3=(2,1,2)$ to obtain orthonormal basis

10. Mention any three properties of Cumulative distribution function

11. Explain briefly the following random variables. (i) Chi-Square Random Variable (ii) Student's t Random Variable

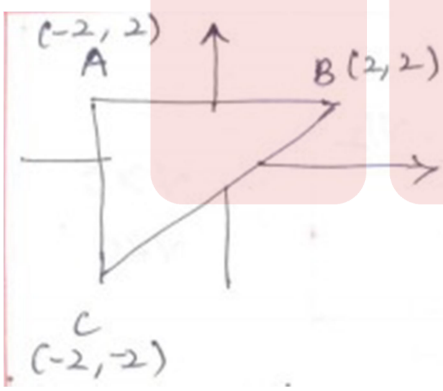
12. The joint PDF $f_{XY}(x,y) = C$ is a constant when (0< $x < 3)$ and $(0 < y < 3)$ and is 0 otherwise.

a. What is the value of the constant C?
b. What are PDFs for X and Y?
c. What is $F_{xy}(x,y)$ when (0< $x < 3)$ and $(0 < y < 3)$?
d. What are $F_{xy}(x,\infty)$ and $F_{xy}(\infty, y)$?

Are X and Y independent

13. Mean and variance of random variable X are -2 and 3. Mean and variance of Y are 3 and 5. The correlation E[XY]= - 8.7. What are the covariance COV[XY] and correlation coefficient ρ_{XY}.

14. Shown in Fig is a region in the x, y plane where the bivariate pdf$f_{XY}(x, y) = c$, elsewhere, the pdf is 0,



What value must c haveEvaluate FXY(1,1).

Find the pdfs of X and Y.

15. Define Central Limit Theorem and show that the sum of the two independent Gaussian random variables is also Gaussian

16. Define the autocorrelation function (ACF) of a random process and discuss its properties

17. With the help of an example, define Random Process and discussdistributions and density functions of a random process.

18. A random Process is defined by $Y(t)=ACos(\omega_c t+\Theta)$ where A and ω_c are constants, but Θ is a random variable distributed uniformly between +/- π.Determine Autocorrelation function $R_y(\tau)$. Pdf of random variable Θ is $1/2\pi$

19. Two zero mean Gaussian random variables X and Y are added. If W=X+Y their variances 3 and 7 respectively. Find the characteristic function of W

20. Describe the column space and the null space of the following matrices.